THE TOP DIFFERENTIAL OF THE HODGE-TO-DE RHAM PAGE VANISHES FOR A SMOOTH PROPER EQUIDIMENSIONAL SCHEME OVER A FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file derham.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-derham/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The bottom part degeneration and the coherent duality remark, and why they do not suffice. Same chapter derham.tex, label lemma-bottom-part-degenerates (cited at proof line 4588) and Duality for Schemes, remark duality-remark-coherent-duality-proper-over-field (cited at proof line 4592). The proof opens by explaining that the tempting duality argument fails because the maps $\mathcal{O}_X \to \Omega^1_{X/k}$ and $\Omega^{d-1}_{X/k} \to \Omega^d_{X/k}$ are not $\mathcal{O}_X$-linear, so functoriality cannot identify the two maps as duals.
- Smoothness, properness and the identification of the base field with the global functions. Varieties, labels varieties-lemma-smooth-geometrically-normal (invoked at proof line 4600) and varieties-lemma-proper-geometrically-reduced-global-sections (invoked at proof line 4601), Algebra, label algebra-lemma-characterize-separable-algebraic-field-extensions (invoked at proof line 4604) and Morphisms, label morphisms-lemma-triangle-differentials (invoked at proof line 4607). They reduce to an irreducible X with $H^0(X, \mathcal{O}_X) = k$, replacing k by a finite separable extension and using $\Omega_{k'/k} = 0$.
- Hodge duality in the top degree. Same chapter derham.tex, label lemma-duality-hodge (invoked at proof lines 4612 and 4642). It gives $\dim H^d(X, \Omega^d_{X/k}) = 1$ and shows that the corresponding group for $X^{(p)}$ is nonzero, which is what makes the surjectivity argument for Θ^d sufficient.

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- The relative Frobenius in characteristic p and the Garel maps attached to it. Same chapter derham.tex, label lemma-Garel-map-frobenius-smooth-char-p (invoked at proof line 4616). It provides the properties of $F_{X/k} : X \to X^{(p)}$ and of the maps Θ^{d-1}, Θ^d used in the commutative diagram; in particular $\Theta^{d-1} = 0$ and Θ^d is surjective.
- Finiteness of the Frobenius, right exactness in top degree, and the spreading out of a smooth proper family. Cohomology of Schemes, labels coherent-lemma-relative-affine-cohomology (invoked at proof line 4629) and coherent-lemma-proper-over-affine-cohomology-finite (invoked at proof line 4662); Cohomology, label cohomology-proposition-vanishing-Noetherian (invoked at proof line 4640); Limits, labels limits-lemma-descend-finite-presentation, limits-lemma-descend-smooth, limits-lemma-descend-dimension-d and limits-lemma-eventually-proper (invoked at proof lines 4658-4660) and limits-lemma-higher-direct-images-zero-above-dimension-fibre (invoked at proof line 4681); Cohomology of Schemes, label coherent-lemma-flat-base-change-cohomology (invoked at proof line 4667). They make the horizontal arrows isomorphisms (finiteness of F), make H^d right exact (vanishing above d), spread X out to a smooth proper family over a finite type $\mathbf{Z}$-algebra, make its cohomology finite free after localization and identify the special fibres by flat base change.

Source reference. Original lemma 090W: sources/stacks-chow/tags.txt, source lines 11424–11424. Navigation pointer for \ref{algebra-lemma-characterize-separable-algebraic-field-extensions} used inside the counted span (line 4605); the argument that consumes it is retained in the span.

Source reference. Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Navigation pointer for \ref{coherent-lemma-flat-base-change-cohomology} used inside the counted span (line 4668); the argument that consumes it is retained in the span.

Source reference. Original lemma 02O6: sources/stacks-chow/tags.txt, source lines 3505–3505. Navigation pointer for \ref{coherent-lemma-proper-over-affine-cohomology-finite} used inside the counted span (line 4663); the argument that consumes it is retained in the span.

Source reference. Original lemma 089W: sources/stacks-chow/tags.txt, source lines 10511–10511. Navigation pointer for \ref{coherent-lemma-relative-affine-cohomology} used inside the counted span (line 4629); the argument that consumes it is retained in the span.

Source reference. Original proposition 02UZ: sources/stacks-chow/tags.txt, source lines 3750–3750. Navigation pointer for \ref{cohomology-proposition-vanishing-Noetherian} used inside the counted span (line 4640); the argument that consumes it is retained in the span.

Source reference. Original remark 0FVX: sources/stacks-chow/tags.txt, source lines 19827–19827. Navigation pointer for \ref{duality-remark-coherent-duality-proper-over-field} used inside the counted span (line 4593); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FW2: sources/stacks-derham/pinned-original-source.tex, source lines 4368–4368. Navigation pointer for \ref{lemma-Garel-map-frobenius-smooth-char-p} used inside the counted span (line 4616); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FW5: sources/stacks-derham/pinned-original-source.tex, source lines 4548–4548. Navigation pointer for \ref{lemma-bottom-part-degenerates} used inside the counted span (line 4588); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FW4: sources/stacks-derham/pinned-original-source.tex, source lines 4496–4496. Navigation pointer for \ref{lemma-duality-hodge} used inside the counted span (line 4588); the argument that consumes it is retained in the span.

Source reference. Original lemma 0EY2: sources/stacks-chow/tags.txt, source lines 18677–18677. Navigation pointer for \ref{limits-lemma-descend-dimension-d} used inside the counted span (line 4659); the argument that consumes it is retained in the span.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Navigation pointer for \ref{limits-lemma-descend-finite-presentation} used inside the counted span (line 4658); the argument that consumes it is retained in the span.

Source reference. Original lemma 0C0C: sources/stacks-chow/tags.txt, source lines 15082–15082. Navigation pointer for \ref{limits-lemma-descend-smooth} used inside the counted span (line 4659); the argument that consumes it is retained in the span.

Source reference. Original lemma 081F: sources/stacks-chow/tags.txt, source lines 10215–10215. Navigation pointer for \ref{limits-lemma-eventually-proper} used inside the counted span (line 4660); the argument that consumes it is retained in the span.

Source reference. Original lemma 0E7D: sources/stacks-chow/tags.txt, source lines 17778–17778. Navigation pointer for \ref{limits-lemma-higher-direct-images-zero-above-dimension-fibre} used inside the counted span (line 4682); the argument that consumes it is retained in the span.

Source reference. Original lemma 01UX: sources/stacks-chow/tags.txt, source lines 2426–2426. Navigation pointer for \ref{morphisms-lemma-triangle-differentials} used inside the counted span (line 4607); the argument that consumes it is retained in the span.

Source reference. Original lemma 0BUG: sources/stacks-chow/tags.txt, source lines 14874–14874. Navigation pointer for \ref{varieties-lemma-proper-geometrically-reduced-global-sections} used inside the counted span (line 4601); the argument that consumes it is retained in the span.

Source reference. Original lemma 056T: sources/stacks-chow/tags.txt, source lines 6721–6721. Navigation pointer for \ref{varieties-lemma-smooth-geometrically-normal} used inside the counted span (line 4600); the argument that consumes it is retained in the span.

Lemma 0.1. *Let k be a field. Let X be a smooth proper scheme over k equidimensional of dimension d. The map*

$$d : H^d(X, \Omega^{d-1}_{X/k}) \to H^d(X, \Omega^d_{X/k})$$

is zero.

Proof. It is tempting to think this follows from a combination of Lemmas 0FW5 and 0FW4. However this doesn't work because the maps $\mathcal{O}_X \to \Omega^1_{X/k}$ and $\Omega^{d-1}_{X/k} \to \Omega^d_{X/k}$ are not $\mathcal{O}_X$-linear and hence we cannot use the functoriality discussed in

Duality for Schemes, Remark 0FVX to conclude the map in Lemma 0FW5 is dual to the one in this lemma.

We may replace X by a connected component of X. Hence we may assume X is irreducible. By Varieties, Lemmas 056T and 0BUG we see that $k' = H^0(X, \mathcal{O}_X)$ is a finite separable extension k'/k. Since $\Omega_{k'/k} = 0$ (see for example Algebra, Lemma 090W) we see that $\Omega_{X/k} = \Omega_{X/k'}$ (see Morphisms, Lemma 01UX). Thus we may replace k by k' and assume that $H^0(X, \mathcal{O}_X) = k$.

Assume $H^0(X, \mathcal{O}_X) = k$. We conclude that $\dim H^d(X, \Omega^d_{X/k}) = 1$ by Lemma 0FW4. Assume first that the characteristic of k is a prime number p. Denote $F_{X/k} : X \to X^{(p)}$ the relative Frobenius of X over k; please keep in mind the facts proved about this morphism in Lemma 0FW2. Consider the commutative diagram

$$\begin{array}{ccccc} H^d(X, \Omega^{d-1}_{X/k}) & \longrightarrow & H^d(X^{(p)}, F_{X/k,*}\Omega^{d-1}_{X/k}) & \xrightarrow{\Theta^{d-1}} & H^d(X^{(p)}, \Omega^{d-1}_{X^{(p)}/k}) \\ \downarrow & & \downarrow & & \downarrow \\ H^d(X, \Omega^d_{X/k}) & \longrightarrow & H^d(X^{(p)}, F_{X/k,*}\Omega^d_{X/k}) & \xrightarrow{\Theta^d} & H^d(X^{(p)}, \Omega^d_{X^{(p)}/k}) \end{array}$$

The left two horizontal arrows are isomorphisms as $F_{X/k}$ is finite, see Cohomology of Schemes, Lemma 089W. The right square commutes as $\Theta_{X^{(p)}/X}$ is a morphism of complexes and Θ^{d-1} is zero. Thus it suffices to show that Θ^d is nonzero (because the dimension of the source of the map Θ^d is 1 by the discussion above). However, we know that

$$\Theta^d : F_{X/k,*}\Omega^d_{X/k} \to \Omega^d_{X^{(p)}/k}$$

is surjective and hence surjective after applying the right exact functor $H^d(X^{(p)}, -)$ (right exactness by the vanishing of cohomology beyond d as follows from Cohomology, Proposition 02UZ). Finally, $H^d(X^{(d)}, \Omega^d_{X^{(d)}/k})$ is nonzero for example because it is dual to $H^0(X^{(d)}, \mathcal{O}_{X^{(p)}})$ by Lemma 0FW4 applied to $X^{(p)}$ over k. This finishes the proof in this case.

Finally, assume the characteristic of k is 0. We can write k as the filtered colimit of its finite type $\mathbf{Z}$-subalgebras R. For one of these we can find a cartesian diagram of schemes

$$\begin{array}{ccc} X & \longrightarrow & Y \\ \downarrow & & \downarrow \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(R) \end{array}$$

such that $Y \to \mathrm{Spec}(R)$ is smooth of relative dimension d and proper. See Limits, Lemmas 01ZM, 0C0C, 0EY2, and 081F. The modules $M^{i,j} = H^j(Y, \Omega^i_{Y/R})$ are finite R-modules, see Cohomology of Schemes, Lemma 02O6. Thus after replacing R by a localization we may assume all of these modules are finite free. We have $M^{i,j} \otimes_R k = H^j(X, \Omega^i_{X/k})$ by flat base change (Cohomology of Schemes, Lemma 02KH). Thus it suffices to show that $M^{d-1,d} \to M^{d,d}$ is zero. This is a map of finite free modules over a domain, hence it suffices to find a dense set of primes $\mathfrak{p} \subset R$ such that after tensoring with $\kappa(\mathfrak{p})$ we get zero. Since R is of finite type over $\mathbf{Z}$, we can take the collection of primes $\mathfrak{p}$ whose residue field has positive characteristic (details omitted). Observe that

$$M^{d-1,d} \otimes_R \kappa(\mathfrak{p}) = H^d(Y_{\kappa(\mathfrak{p})}, \Omega^{d-1}_{Y_{\kappa(\mathfrak{p})}/\kappa(\mathfrak{p})})$$

for example by Limits, Lemma 0E7D. Similarly for $M^{d,d}$. Thus we see that $M^{d-1,d} \otimes_R \kappa(\mathfrak{p}) \to M^{d,d} \otimes_R \kappa(\mathfrak{p})$ is zero by the case of positive characteristic handled above. $\square$